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India SMART UTILITY Week 2026

10 – 14 March 2026, New Delhi

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Session : Evolving Architecture of the 21st Century Grid with Two Way Power Flows

Advanced Network Management System for OT: A Unified Approach to Digital Utility Operation

Presented By

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INTRODUCTION

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Title: Advanced Network Management System for OT:*A Unified Approach to Digital Utility Operation*

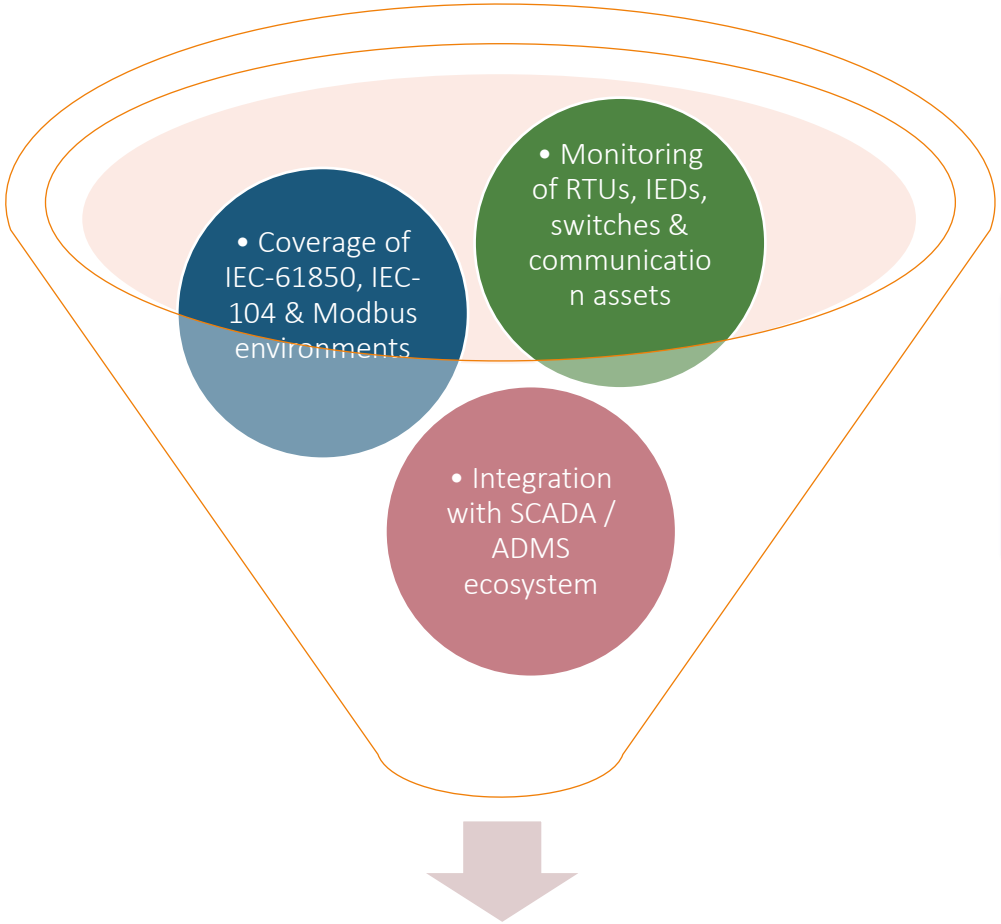
Objective

- Enhance visibility, reliability & cybersecurity of OT networks

- **Enable centralized health monitoring of OT infrastructure**

- **Transition from reactive to predictive grid operations**

Scope



Implementation Features

Real-Time Intrusion Detection & Threat Monitoring
- Continuous OT Network and Asset Surveillance
- Deep Packet Inspection (DPI) for OT Protocols.

Centralized Monitoring & Security Analytics
- Unified Dashboard for Security Insights
- Historical Event Analysis

Asset Management & Lifecycle Optimization
- Asset Life prediction and tracks asset degradation.
- Asset maintenance priority schedule.

BACKGROUND

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Background & Challenges

Volume

Tata Power Grids: 100, Remote Terminal Units(RTUs): 160, Protection Relay IEDs: 9000, Power Transformer 230,Network & Security Devices

Infrastructure

No process for asset management, Fragmented visibility across IEDs, RTUs, and network devices, Required Grid Operation Reliability Excellence.

Spread

Extensive Coverage over 510 Sq. Km Area, Lack of Advanced OT devices Performance & Security Monitoring, Expanding EV Infrastructure

Challenge

Scalability limitations, absence of predictive fault and threat detection, expanding attack surface, geographically complex operations, and high reliance on manual field response.

Key Highlights of the project

Asset Management and Health Monitoring

- Centralized asset inventory
- Real-time asset health and network monitoring
- AI/ML-based detection of asset critical alarms and traffic anomalies
- Preventive to Predictive Maintenance

Enhanced Grid Reliability and Resilience

- Advanced OT protocol monitoring (IEC 104, IEC 61850-MMS & GOOSE) detects anomalies.
- Unauthorized login devices and cyber threats and ensure network stability.

Compliance-Driven Predictive & Secure Operation

- Supports regulatory compliance while enhancing operational efficiency through predictive analytics
- Real-time traffic behavior monitoring
- Early detection of performance degradation and abnormal conditions.

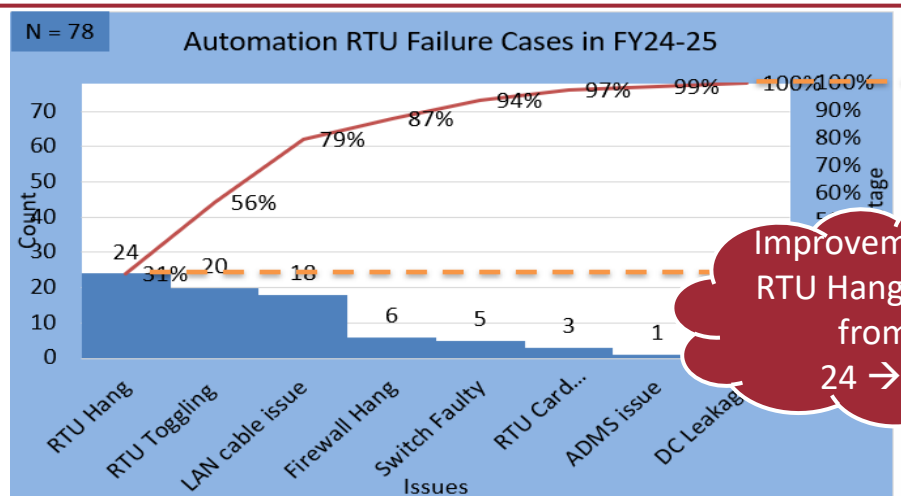
RELEVANCE

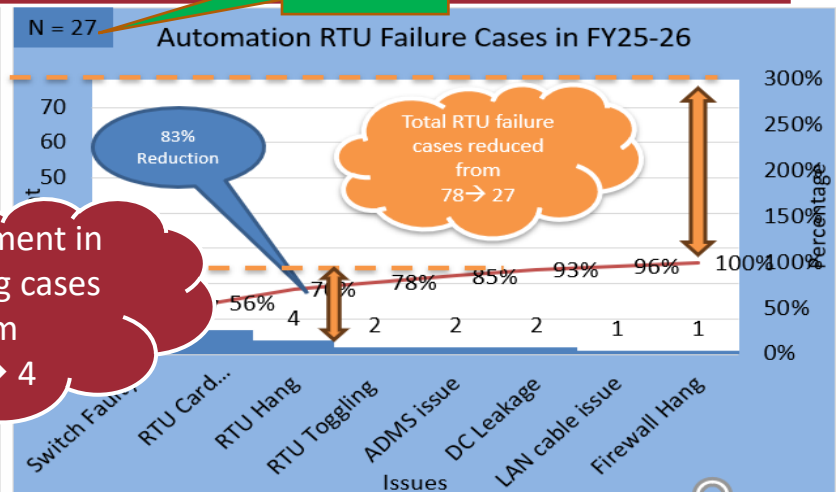
Enhanced Asset and Network Visibility:
Enables OT asset discovery, communication mapping, and detection of unauthorized devices and traffic anomalies.

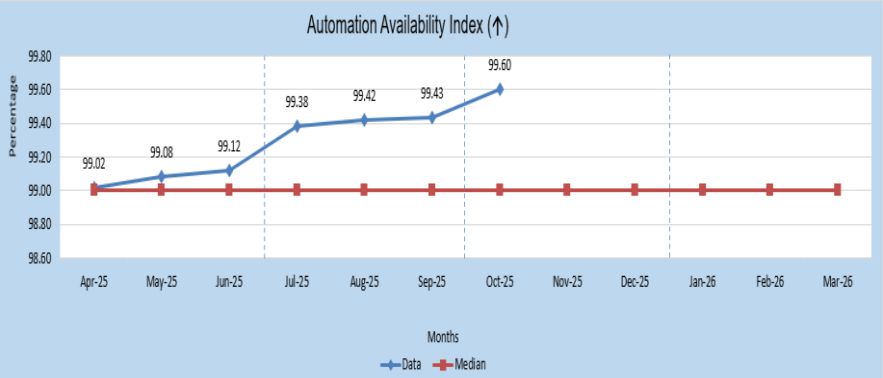
Improved Network Resilience:
Ccontinuous traffic monitoring with real-time anomaly detection and proactive management.

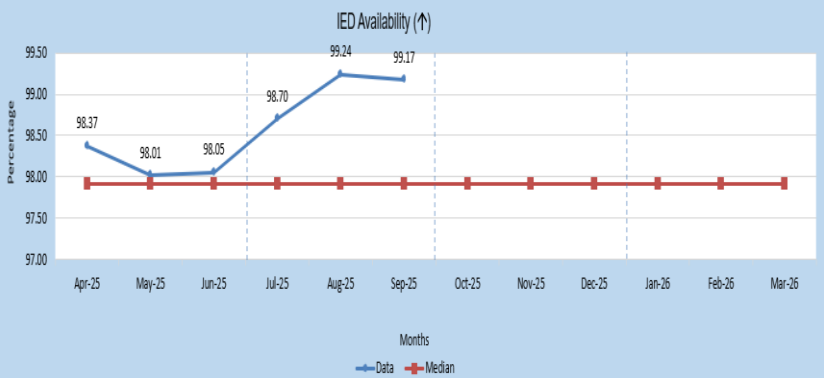
Improvement in Operational Efficiency: The proactive approach helped in improving Operational Efficiency and Reliability across the Grid distribution network.

Customer Satisfaction: Customer complaints related to Grid Communication failure has significantly reduced improving overall experience of the customer









- **Significant 65% reduction in Grid Automation RTU failures.**
- **Preventive to Predictive maintenance shift, reducing repeat failures and emergency field visits.**
- **Early detection of link and node degradation enabled proactive intervention, resulting in 20–30% reduction in SCADA communication downtime and Improved grid availability and operational continuity.**

OT-NMS IMPLEMENTATION

Tata Power DDL - NMS

Fri 12/5/2025 3:04 PM

ON

OT NMS

OT-NMS Alert

To Ankur Kaushik

Cc Abhinav Mogha

```
##################################################################
Group Name: OT-NMS-FW
Responsible_Person: Ankur Kaushik
##################################################################
ALERT SUMMARY:
Node Down: 1[redacted](RHRD-33KV-FW-CHECKPOINT-1200R)
------------------------------------------------------------------
Other Details:
SOURCE NODE:RHRD-33KV-FW-CHECKPOINT-1200R
Management Address IP: 1[redacted]
SEVERITY:Critical
STATE: ot-nms
if you have any queries regarding this email, then contact OT-NMS Administrator
------------------------------------------------------------------

Regards,
OT-NMS Team
```

Grid Name							Uptime	First Seen	Last Seen
NMS server	Datacenter						0 y 1 m 4 d 17 h 57 mn	2025-05-27 13:57:04	2025-12-05 15:03:05
RANIBAGH	Buscoupler						0 y 1 m 26 d 17 h 14 mn	2025-05-27 14:53:01	2025-12-05 15:04:15
WAZIRABAD	RTU Panel						0 y 3 m 27 d 20 h 50 mn	2025-05-29 15:50:42	2025-12-05 15:04:29
AZADPUR	Control Room						1 y 4 m 10 d 1 h 18 mn	2025-06-06 10:13:46	2025-11-07 12:28:26
BAWANA_CLEAR_WATER	RTU Panel						0 y 0 m 21 d 21 h 2 mn	2025-06-06 10:14:02	2025-12-05 15:04:25
0 BHALSWA	RTU Panel						0 y 0 m 6 d 22 h 9 mn	2025-06-06 10:14:04	2025-12-05 14:54:56
RG-4	In RTU PanelCPU B						0 y 0 m 8 d 4 h 8 mn	2025-06-20 12:40:15	2025-12-05 15:02:44
ROHINI_23	SCADA ROOM						0 y 0 m 6 d 21 h 58 mn	2025-06-20 12:40:49	2025-12-05 14:56:00
ROHINI_23	SCADA ROOM						0 y 0 m 6 d 22 h 5 mn	2025-06-20 12:41:26	2025-12-05 15:03:02
ROHINI_23	RTU Panel						0 y 1 m 19 d 3 h 17 mn	2025-06-20 12:43:09	2025-12-05 14:58:40
ROHINI_34	RTU Panel						0 y 11 m 26 d 4 h 14 mn	2025-06-20 13:16:42	2025-12-05 15:04:45
RANIBAGH CC	RTU Panel						0 y 0 m 23 d 2 h 22 mn	2025-06-20 13:18:05	2025-12-05 14:55:54
SHAKTI NAGAR	RTU Panel						0 y 0 m 20 d 23 h 41 mn	2025-06-20 13:26:29	2025-12-05 18:01:36
RANIBAGH							0 y 2 m 29 d 6 h 10 mn	2025-06-20 13:40:55	2025-12-05 18:01:46
RG-4	In RTU Panel CPU A						0 y 0 m 8 d 4 h 11 mn	2025-06-23 11:53:58	2025-12-05 15:02:58
BAWANA_4	11KV BUS PT-3						0 y 3 m 8 d 18 h 55 mn	2025-06-23 14:21:15	2025-12-05 18:03:18
DHEERPUR	In SCADA Room CP						0 y 0 m 4 d 0 h 14 mn	2025-06-26 12:20:40	2025-12-05 14:55:49
DHEERPUR	In SCADA Room CP						0 y 1 m 12 d 11 h 22 mn	2025-06-26 12:21:46	2025-12-05 14:55:55
BAWANA_CLEAR_WATER	Control Room						0 y 0 m 21 d 20 h 56 mn	2025-06-30 12:00:12	2025-12-05 14:55:29
ROHINI_22	In RTU Panel		Synergy	Husky		Up	0 y 0 m 6 d 12 h 35 mn	2025-06-30 12:49:47	2025-12-05 18:02:50
ROHINI_22	In RTU Panel		Synergy	Husky		Down	0 y 1 m 6 d 1 h 48 mn	2025-06-30 12:49:49	2025-12-05 14:58:25
Azadpur	Control Room		GE	D400		Up	0 hrs	2025-07-03 16:38:32	2025-12-05 18:00:18
BCC	Datacenter		GE	D400		Down	0 hrs	2025-07-03 17:10:26	2025-12-05 15:04:29
MCC	Datacenter		GE	G500		Down	0 hrs	2025-07-03 17:10:51	2025-12-05 14:57:02
BCC	Datacenter		GE	D400		Down	0 hrs	2025-07-03 17:11:19	2025-12-05 14:57:28
BCC	Data Center	X21021YPKTF8Y8C	Sophos	Sophos XG210	20.0.2 MR-2-Build378	Down	0 y 11 m 2 d 6 h 12 mn	2025-07-03 17:13:39	2025-12-05 15:02:45
MCC	Data Center	X210218T8DQ8C3A	Sophos	Sophos XG210	20.0.2 MR-2-Build378	Down	0 y 5 m 5 d 19 h 34 mn	2025-07-03 17:15:10	2025-12-05 14:58:35
MCC	Data Center	X21021833VKVJ93	Sophos	Sophos XG210	20.0.2 MR-2-Build378	Down	0 y 8 m 3 d 12 h 18 mn	2025-07-03 17:16:12	2025-12-05 14:57:39
MCC	Datacenter		GE	D400		Down	0 hrs	2025-07-03 17:17:35	2025-12-05 15:04:19

Scalability and Replication Potential of the Project

Tata Power-DDL Achieved Target

2,000,000 customers

100 Grid Stations

Integrating
10000 devices

- 160 RTUs
- 9000 IEDs
- 230 PTRs
- 515 Switches
- 95 Firewalls

The Next-Gen OT-NMS Implementation makes it an **ideal candidate for integration** into **other Tata Discoms and 220KV DTL Grids** , enabling seamless deployment across **diverse geographic locations and infrastructures**.

Next-Gen OT-NMS ensures **scalable security** for Smart Grid rollout, Smart Meters, DERs, EV charging, and IED integration, enhancing **resilience, threat detection and operational efficiency** across the utility sector.

Future Next-Gen OT-NMS Integration:

- **OT network digital twins to simulate configuration changes, firmware upgrades, redundancy behavior, and traffic patterns before live deployment, minimizing operational risk and outages.**
- **Historical OT telemetry, protocol trend analysis, and configuration drift detection to support risk-based maintenance and operational planning.**
- **AI-driven predictive analytics to forecast device degradation, communication bottlenecks, protocol instability and proactive grid reliability management.**

OT-NMS Financial Benefits

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Automation Asset Failure Cost Reduction

- Automation asset failures incurred ₹1.15 Cr during FY23–FY25 across critical assets such as CPU cards, Ethernet switches, firewalls, and communication modules at grid locations.
- OT-NMS reduces failures by 65% in FY26, saving ₹25 Lakhs.

Manpower & Vehicle Cost Reduction

- FY25–26 recorded 256 field incidents (78 RTU breakdowns + 178 OT network hang cases), each requiring 2 automation personnel and 1 vehicle for on-site restoration.
- Avoided manpower and vehicle deployment results in annual cost reduction of ₹19.2 Lakhs.

SAIDI Improvement & Cost Reduction

- STS SAIDI improved from 0.83 hrs (FY24–25) to 0.65 hrs (FY25–26), achieving 0.18 hrs (10.8 minutes) reliability gain.
- SAIDI improvement results in annual benefit of ₹17.05 Lakhs.

Projected Benefit with OT-NMS	
Item	**Value**
Historical Failure Cost (FY23–FY25)	₹1.15 Cr
Failure Reduction Assumed	65%
Estimated Cost Saved per year	₹25 Lakhs

Annual Manpower Cost Saving – OT Network	
Item	**Value**
Total Incidents (Yearly)	256
Personnel per Incident	2
Cost / Person / Site	₹3,000
Vehicle Cost / Site	₹1,500
Cost per Incident	₹7,500
Manpower Cost / Year	₹15.36 Lakhs
Vehicle Cost / Year	₹3.84 Lakhs
Total Annual Cost Saving	₹19.2 Lakhs

SAIDI Improvement Calculation with MU Served	
Item	**Value**
Total MUs served by TPDDL in a year	10000
Total Units served by TPDDL in a minute	19025.88
Improvement in SAIDI STS in Hrs w.r.t FY 25-26	0.18
Improvement SAIDI STS in Minutes w.r.t FY 25-26	10.8
Total Units served by TPDDL in 10.8 minutes	205479.45
Average Billing Rate	₹ 8.30
Total Benefit in a Year	₹ 17 Lakhs

Total Cost Saving Per year

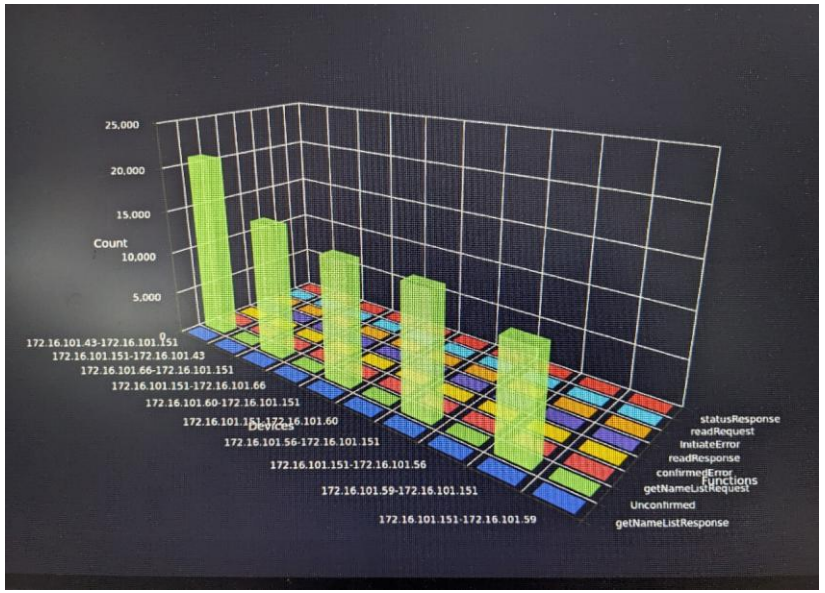
0.61 Cr

Business Case Study :

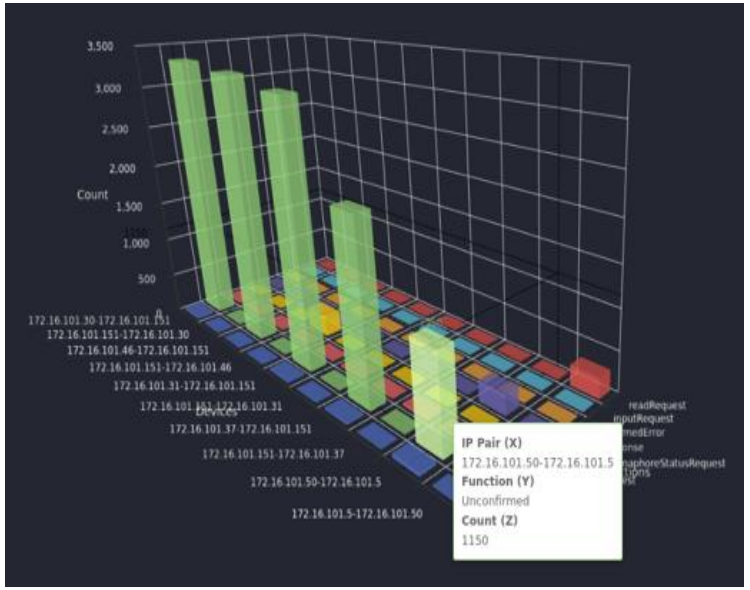
Jahangirpuri Grid -

Unconfirmed data exchange (measurements) between RTU and IEDs(LDR) observed – 15k to 20k count

Using Next-Gen OT-NMS Deep Packet Inspection tools statistics helped us to clearly identify the root cause of the problem.



Relay 61850 Datasets modified



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THANK YOU

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